

Session 04 :

The Strategic Role of Information Systems



MGT 3123: Management Information System
Department of Management and Entrepreneurship
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What is Information System?

An Information System is a **combination** of *people, technology, data, and processes* that collects, stores, processes, and distributes information to support organisational operations and decision-making.

Components of Information Systems

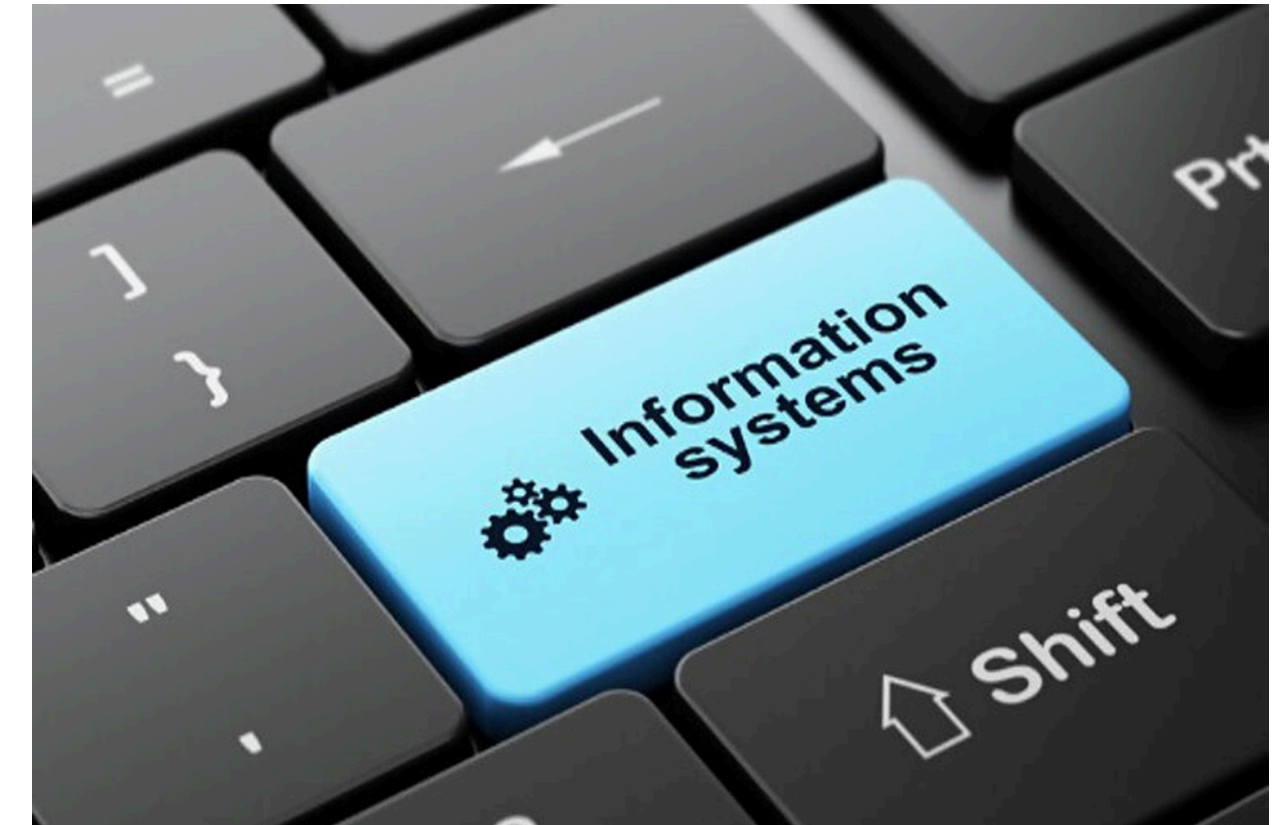
Hardware – computers, servers, devices

Software – applications, databases

Data – facts and figures

People – users, managers, IT staff

Processes – procedures for collecting and using data



What is Strategy?

Strategy is about **making choices and creating a unique position** by performing different activities from rivals or performing similar activities in a different way.

What is Strategic role?

The way an **activity, resource, or system contributes** to achieving an organization's **long-term goals, sustainable competitive advantage,** and overall direction

How something helps the organization win in the long run.



Competitive Advantages

Competitive advantage is the ability of a company to **perform better than its competitors** by **offering greater value** through *lower prices, better products, or unique customer experiences*.

- Cost Leadership
- Differentiation
- Internal Resources
- Capabilities



Sustainable Competitive Advantage (SCA)



Why Information Systems Matter Strategically

- Improve efficiency & productivity
- Enable better decision-making
- Create competitive advantages
- Transform business models (e.g., Uber, Airbnb)
- Support globalisation



The Evolution of IS – From Back-Office to Boardroom

- The Changing Perception:
- 1960s-70s: Data Processing – Transaction automation, cost reduction.
- 1980s: Management Reporting – MIS, DSS for managerial control.
- **1990s: Strategic Weapon – SIS, ERP, the Internet.**
- 2000s: Collaborative Core – Integration, supply chains, CRM.
- 2010s-Present: Digital Foundation – Platform, analytics, AI, ecosystem driver.



Operational, Managerial & Strategic roles of IS

- Operational Role – Routine tasks (payroll, inventory)
- Managerial Role– Monitoring & controlling (MIS, dashboards)
- **Strategic Role– Long-term planning & competitive advantage**
(ERP, AI, Big Data)

Example: Sri Lankan supermarkets using POS + ERP for forecasting.



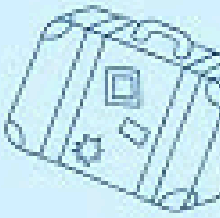
Mandarin Oriental - Customer Relationship Management Systems

Digital Travel
APAC

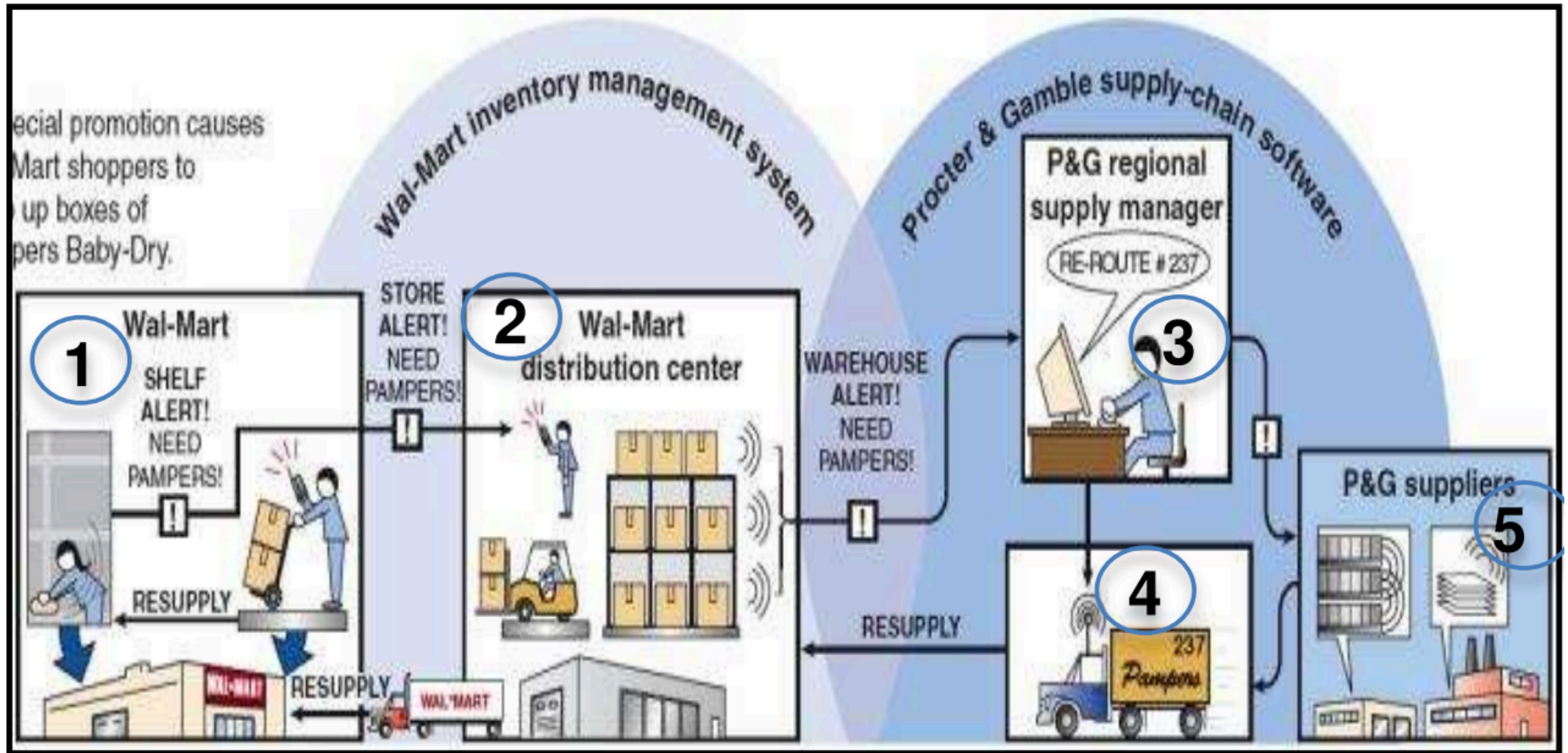
**FROM DATA CHAOS
TO \$1 BILLION:**
How Mandarin Oriental
Make Customers
Come Back More

Alex Schellenberger
Chief Brand &
Marketing Officer


MANDARIN ORIENTAL



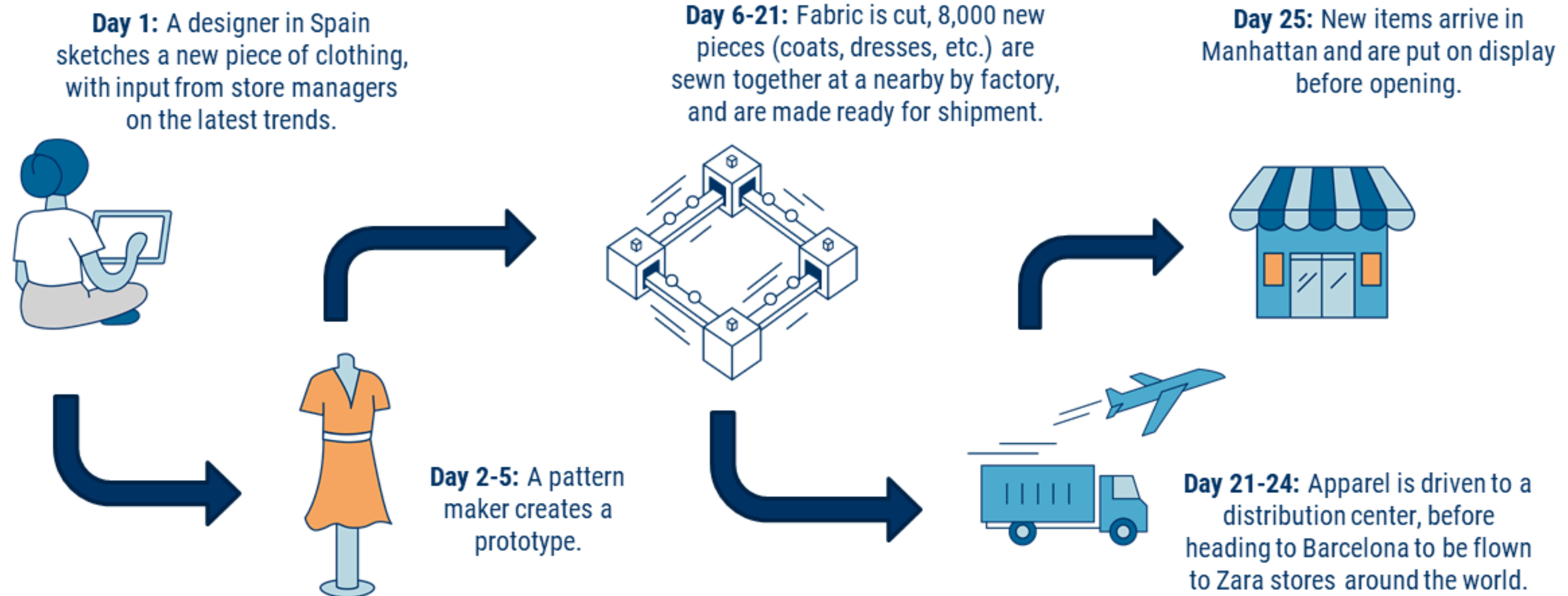
Walmart - Retail Link System



Zara - Real-Time Fashion IS

Fast fashion's speedy supply chain quickly caters to new trends

Fast fashion retailer Zara, owned by Spain-based Inditex, can get a piece of apparel from a design workshop in Spain to a display rack in a Manhattan store in **25 days**.



CBINSIGHTS

Source: [The Wall Street Journal](#)

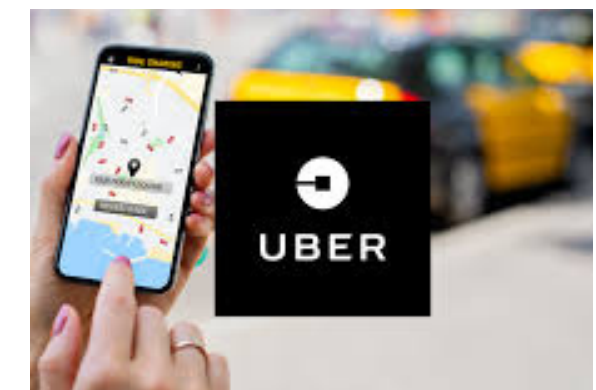


Porter's Five Forces

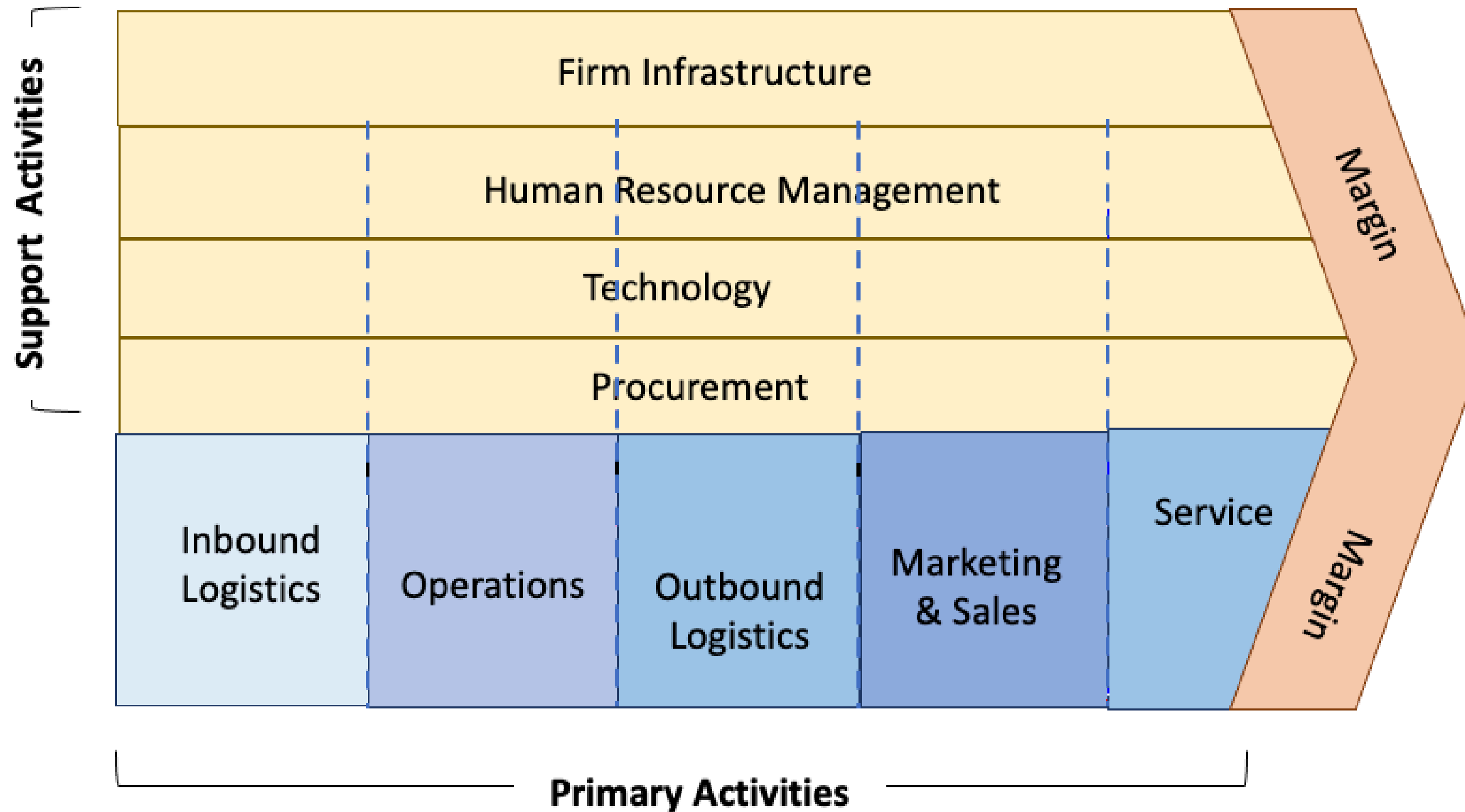


Porter's Five Forces & Information Systems

- IS can reduce competitive pressures by:
 - Lowering buyer power (loyalty apps, personalization)
 - Lowering supplier power (Walmart's SCM)
 - Creating entry barriers (AI-driven operations, Uber)
 - Differentiation (Netflix recommendations)
 - Cost leadership (Toyota's information-driven manufacturing)



Porter's Value Chain



Porter's Value Chain & IS

- IS strengthens all value chain activities:
 - Inbound logistics – RFID, IoT tracking
 - Operations – Automation, robotics
 - Outbound logistics – Route optimization
 - Marketing & Sales – CRM, digital ads
 - Service – Chatbots, online support



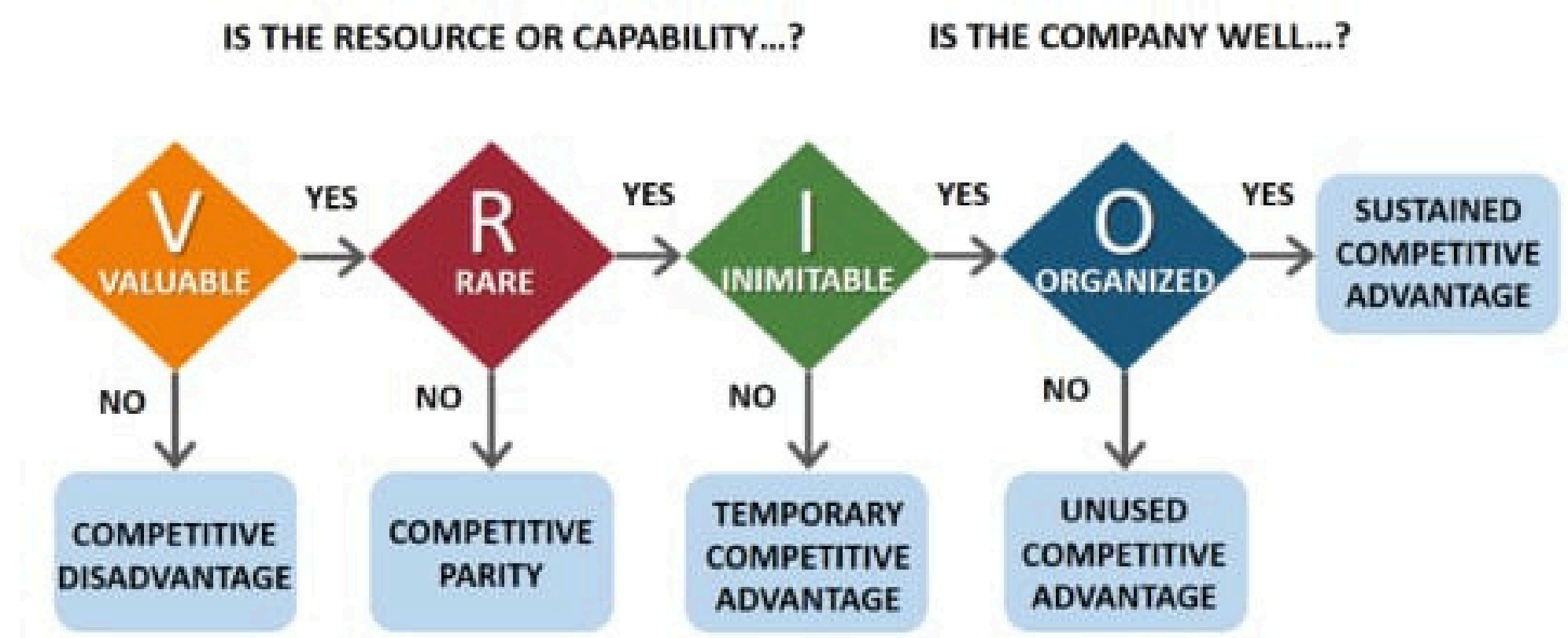
- *Example: ZARA's real-time design–manufacturing system.*



The Resource-Based View (RBV) & IS

- IS as a strategic resource if it is:

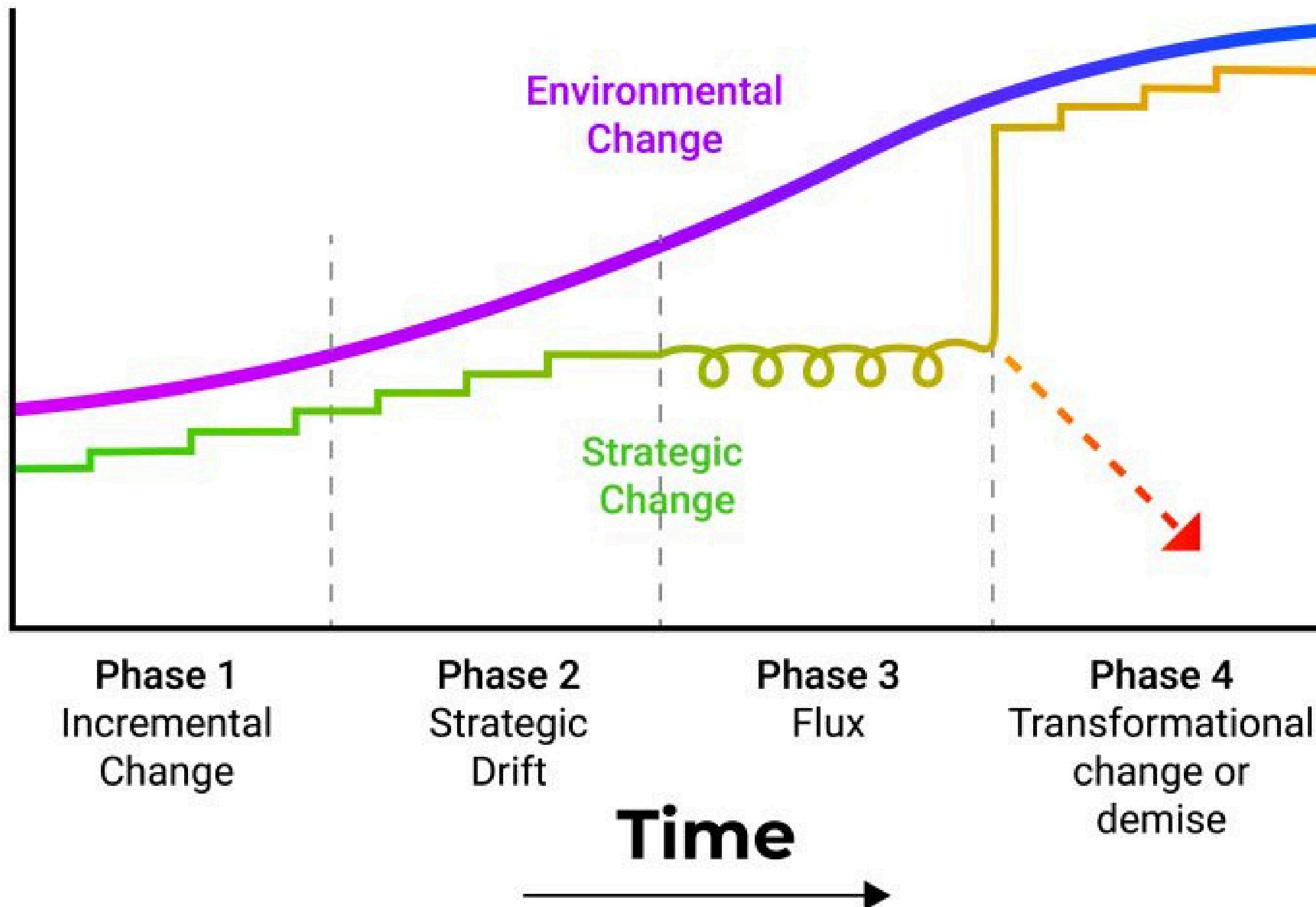
- Valuable
- Rare
- Inimitable
- Organised



- *Example: Netflix's proprietary viewing-data algorithm = high VRIO value.*



Avoid Strategic Drift with IS

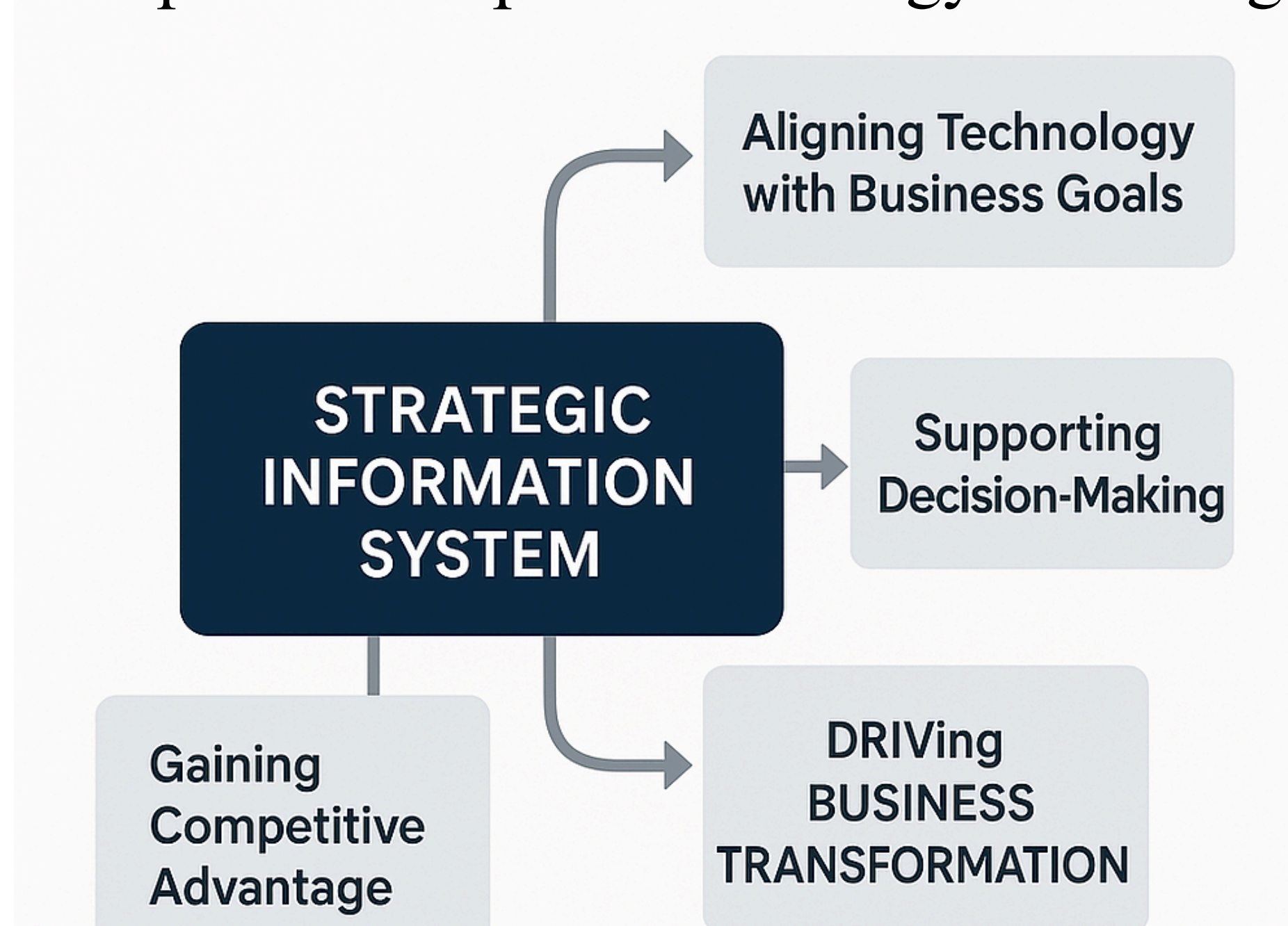


© Johnson, Scholes & Whittington (2011), Exploring Strategy



Strategic Information Systems (SIS)

A Strategic Information System (SIS) is an information system that is developed or used to support or shape the competitive strategy of an organisation.



Strategic Information Systems (SIS) - Cont

Regular IS	SIS
Supports routine operations	Changes the way the organisation competes
Improves efficiency	Creates competitive advantage
Short-term	Long-term
Back-office focus	Customer, supplier & competitive focus
Automates	Innovates
Payroll systems, Attendance systems, Inventory updates	Uber's real-time matching & surge algorithm, Zara's real-time fashion data system, Walmart's RFID-based



Ethics Vs Strategy

Strategy is about achieving business goals, gaining competitive advantage, and ensuring profitability. *It focuses on what a firm does to succeed.*

Ethics is about moral principles, fairness, and responsibility toward stakeholders. *It focuses on how a firm achieves its goals.*

Example :

Strategic goal: Maximize profit by cutting costs.

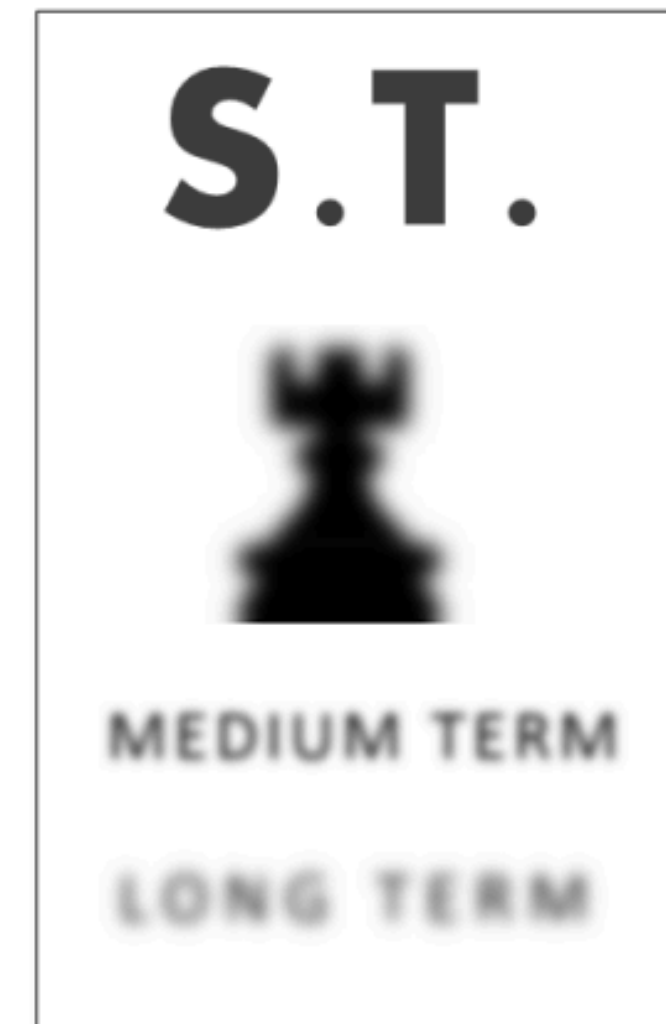
Ethical concern: Laying off workers without proper compensation or ignoring environmental laws.



A purely profit-driven strategy without ethics can lead to short-term gains but long-term reputational and legal risks.

Strategic Myopia:

Can't See Beyond
The Short Term?



Ethical Strategy

An ethical strategy ensures that **how goals are achieved matters** as much as the goals themselves. It aligns *competitive advantage* with:

- *Moral values*
- *Legal compliance*
- *Social and environmental responsibility*



Key Challenges

- **Data Privacy and Confidentiality** - Organizations collect large volumes of user data for analytics and monetization.
- **Surveillance and Employee Monitoring** - Excessive monitoring of employees through IS can violate privacy and autonomy
- **Customer Manipulation**
- **Misuse of Information Systems**
 - Escape Regulations
 - Escape social responsibilities



Volkswagen Emissions Scandal (2015)



Facebook – Cambridge Analytica (2018)



Uber – “Greyball” Program (2017)

Uber used Greyball fake app to evade police across Europe, leak reveals

Firm's bosses knew it was fooling authorities with tool showing phantom cars, documents show



Principles of Ethical Strategy

- **Transparency:** Clear communication with stakeholders.
- **Fairness:** Equitable treatment of employees, customers, suppliers.
- **Accountability:** Accepting responsibility for business outcomes.
- **Sustainability:** Considering long-term social environmental, and economic impact.

